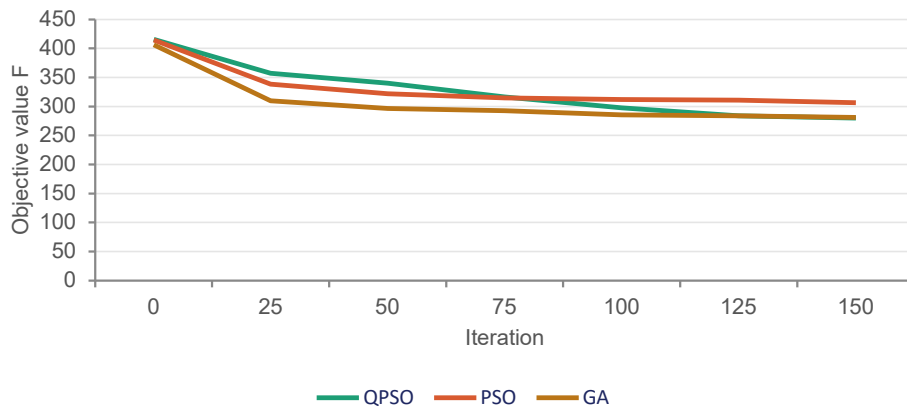


CONVERGENCE ANALYSIS & BENCHMARKING

1 Convergence curve (fitness vs iteration)

Measured across 12 independent runs — 20-node synthetic instance



Metrics reported (per algorithm, per instance)

Distance

Travel time

Congestion

Runtime

Convergence iter.

Quality vs exact

2 Fair benchmarking protocol

- 1 **Population size** — 30 particles/chromosomes — identical across QPSO, PSO, GA
- 2 **Stopping criterion** — 150 iterations, fixed for every algorithm
- 3 **Independent runs** — 12 runs per instance; report mean $\pm$ std dev
- 4 **Hardware** — Same machine, same session, for every algorithm
- 5 **Problem instances** — Same nodes, demand and time windows for all
- 6 **Random seeds** — Fixed seed set, reused across algorithms

Benchmark hierarchy (progressive scale)

Synthetic
10–50 nodes



Public VRP
benchmark



Real OSM
network



Dynamic SUMO
scenario